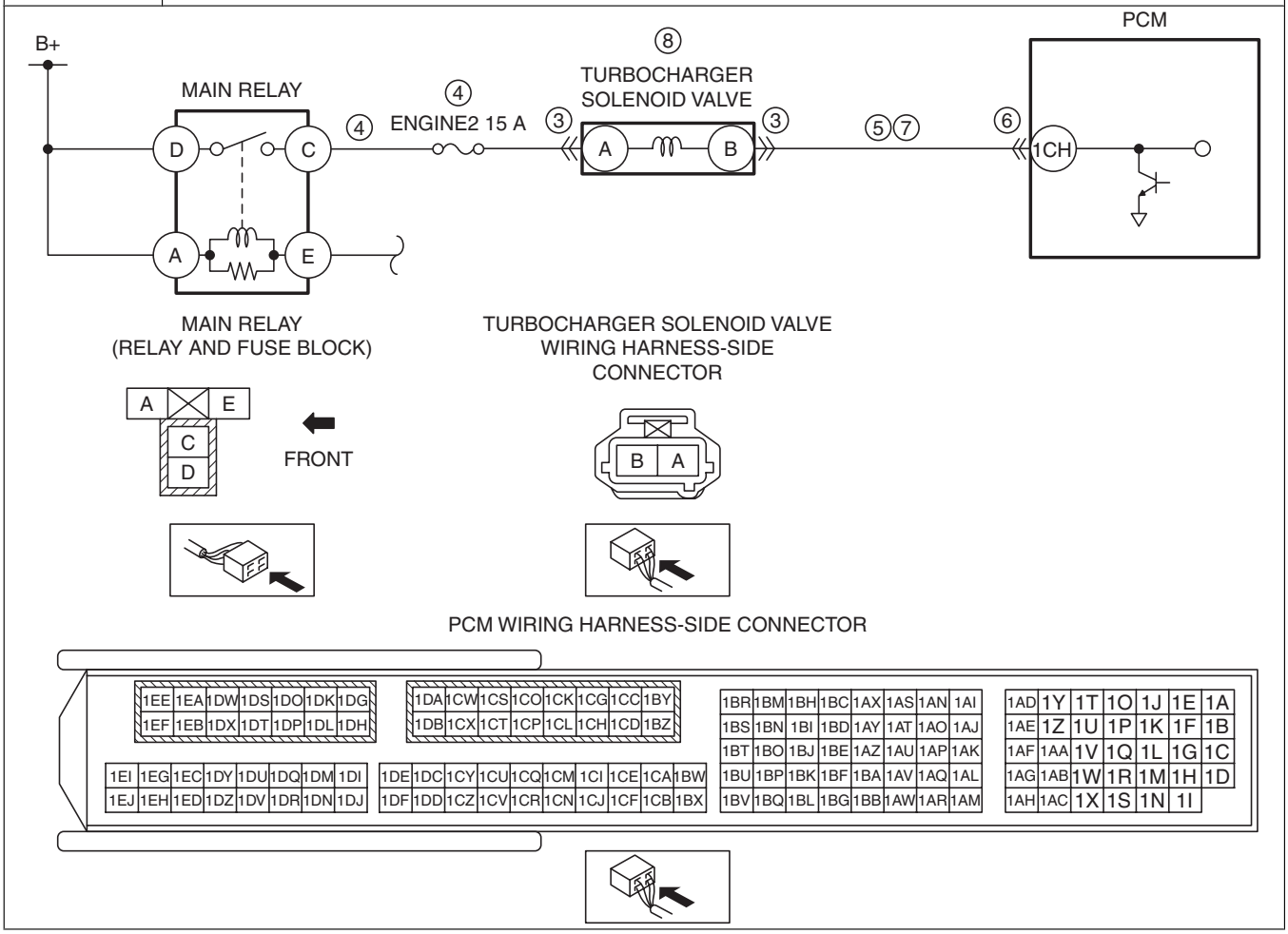
DTC P0047:00 [SKYACTIV-D 1.5]

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DTC P0047:00	Turbocharger solenoid valve control circuit low input
DETECTION CONDITION	• If the PCM detects that the turbocharger solenoid valve voltage at the PCM terminal 1CH is less than the specified value for **1 s** with the following condition met, the PCM determines that the turbocharger solenoid valve circuit voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) **Diagnostic support note** • This is an intermittent monitor (turbo system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the boost control. • Inhibits the EGR control. • PCM restricts engine-transaxle integration control. (ATX) • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Turbocharger solenoid valve connector or terminals malfunction • Short to ground or open circuit in turbocharger solenoid valve power supply circuit — Short to ground in wiring harness between ENGINE2 15 A fuse and turbocharger solenoid valve terminal A — ENGINE2 15 A fuse malfunction — Open circuit in wiring harness between main relay terminal C and turbocharger solenoid valve terminal A • Short to ground in wiring harness between turbocharger solenoid valve terminal B and PCM terminal 1CH • PCM connector or terminals malfunction • Open circuit in wiring harness between turbocharger solenoid valve terminal B and PCM terminal 1CH • Turbocharger solenoid valve malfunction • PCM malfunction

DTC P0047:00	Turbocharger solenoid valve control circuit low input



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT TURBOCHARGER SOLENOID VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the turbocharger solenoid valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT TURBOCHARGER SOLENOID VALVE POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the turbocharger solenoid valve connector is disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the turbocharger solenoid valve terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the ENGINE2 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE2 15 A fuse and turbocharger solenoid valve terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and turbocharger solenoid valve terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
5	**INSPECT TURBOCHARGER SOLENOID VALVE CONTROL CIRCUIT FOR SHORT TO GROUND** • Verify that the turbocharger solenoid valve connector is disconnected. • Switch the ignition off. • Inspect for continuity between turbocharger solenoid valve terminal B (wiring harness-side) and body ground. • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between turbocharger solenoid valve terminal B and PCM terminal 1CH. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 9.
		No	Go to the next step.
6	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.

STEP	INSPECTION		ACTION
7	**INSPECT TURBOCHARGER SOLENOID VALVE CONTROL CIRCUIT FOR OPEN CIRCUIT** • Verify that the turbocharger solenoid valve and PCM connectors are disconnected. • Inspect for continuity between turbocharger solenoid valve terminal B (wiring harness-side) and PCM terminal 1CH (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between turbocharger solenoid valve terminal B and PCM terminal 1CH. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
8	**INSPECT TURBOCHARGER SOLENOID VALVE** • Inspect the turbocharger solenoid valve. (See TURBOCHARGER SOLENOID VALVE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the turbocharger solenoid valve, then go to the next step. (See TURBOCHARGER SOLENOID VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.